26900188 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCECT302 - EMBEDDED SYSTEMS WITH AVR MICROCONTROLLER

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. What does `SBI PORTB, 2` do? [K] / [CO1]

   a). Clears bit 2 of PORTB b). Sets bit 2 of PORTB

   c). Reads bit 2 of PORTB d). Clears PORTB

2. Which interrupt is generated when Timer 0 overflows? [U] / [CO2]

   a). TOV0 b). INT0

   c). USART d). ADC interrupt

3. A stepper motor rotates in: [U] / [CO3]

   a). Continuous random motion b). Discrete angular stepsOnly one direction

   c). Only one direction d). Linear motion only

4. Which assembler directive is used to define a constant value? [K] / [CO1]

a). .ORG b). .EQU

c). .END d). .BYTE

5. How many PORTS available in ATMega32? [K] / [CO2]

a). 1 b). 2

c). 3 d). 4

6. Bluetooth is suitable for: [U] / [CO3]

a). Short-distance data exchange b). Long-distance satellite communication

c). High-voltage switchingAC power transmission d). AC power transmission

7. If register R16 contains 05H and the instruction LDI R16, 0AH is executed, what will be the new content of R16 [A] / [CO1]

a). 05H b). 00H

c). 0FH d). 0AH

8. What is the primary function of the OCR register? [U] / [CO2]

a). Store the current timer count b). Store the value used for comparison with TCNT

c). Select the timer clock d). Enable global interrupts

9. Which AVR port configuration is required when an IR sensor output is connected to an AVR? [U] / [CO3]

a). High impedance outputOutput b). output

c). Input d). Analog output only

10. Which instruction uses immediate addressing? [K] / [CO1]

a). LDI R16, 25 b). MOV R16, R17

c). LD R16, X d). LD R16, Z

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11.	Differntiate between conditional branching and unconditional branching in AVR	[U] / [CO1]
12.	Explain the procedure for generating a time delay using an AVR timer in Normal mode	[A] / [CO2]
13.	How do you configure an AVR port pin as an output to control an LED?	[An] / [CO3]
14.	What are the main components of a microcontroller? Differentiate between a microprocessor and a microcontroller.	[K] / [CO1]
15.	Write an AVR Embedded C program to configure Port B as an output port and send a logic HIGH .	[A] / [CO2]
16.	What is the function of the Enable (EN) pin of an LCD?	[U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17.	Draw and explain the block diagram of the AVR ATmega32 architecture. Describe the CPU, register, ALU, program memory, data memory and peripheral interfaces.	[U] / [CO1]
18.	Explain the memory organization of ATmega32 with a neat diagram. Discuss Program Counter, Program Memory, Data Memory, SRAM, EEPROM, and Stack.	[U] / [CO1]
19.	Describe the steps involved in transmitting a character and explain the role of the UDRE and TXC flags.	[K] / [CO2]
20.	Explain I2C communication in AVR.	[U] / [CO2]

[10x2 = 20]